

ABYLAY AMANBAYEV

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SUMMARY

Pursuing PhD in Computer Science with research focus on Vector Search and Query Optimization.

SKILLS

Languages	Python, SQL, C/C++, HTML/CSS, L ^A T _E X
DS/ML	Pandas, NumPy, PyTorch, TensorFlow, Sklearn, Matplotlib, HuggingFace, LangChain
DBMS	PostgreSQL, MySQL, PG-vector, MilvusDB, Pinecone, FAISS, PASE
Tools	SciPy, BS4, Docker, Git

EDUCATION

University of California Merced Aug 2023 - Present

PhD in Computer Science

- Scientific Advisor: Prof. Florin Rusu
- Courses: LLMs, DL/RL, DBMS implementation, Approximation Algorithms

National Research University Higher School of Economics 2021-2023

Master of Data Science

- Courses: Applied ML, SQL, Computational Complexity, Advanced Algorithms and Data Structures

EXPERIENCE

University of California Merced Aug 2023 - Present

Research/Teaching Assistant

- Research work related to Query Optimization and Vector Databases
- TA for: CSE 111 - Database Systems (Fall 2023 - 2024)
- TA for: CSE 140 - Computer Architecture (Spring 2025)

Lawrence Livermore National Laboratory July-Aug 2024

Team Lead in Data Science Challenge

- Led a team of two undergraduates and one MS student in cardiac research utilizing machine learning methods
- Developed algorithms for cardiovascular diseases diagnostics with over 98% accuracy
- Trained neural networks to reconstruct cardiac activation and transmembrane potential maps from ECG data

RESEARCH

1. Bridging Filtered Vector Search and Information Retrieval via Realistic Dataset (*work in progress*)
2. Sketched Sum-Product Networks for Joins, arXiv:2506.14034 [cs.DB] 16 Jun 2025, B. Tsan, A. Amanbayev, A. Datta, F. Rusu.
3. PLANSIEVE: Real-time Suboptimal Plan Identification through Incremental Learning, arXiv: 2501.16544 [cs.DB] 27 Jan 2025. A. Datta, B. Tsan, Y. Izenov, A. Amanbayev, F. Rusu.
4. Spanning Tree-Based Query Plan Enumeration, arXiv: 2403.04026v1 [cs.DB] 6 Mar 2024. Y. Izenov, A. Datta, B. Tsan, A. Amanbayev, F. Rusu.

AWARDS

1. UC Merced - Blackstone LaunchPad by StartupTree Ideas Competition Winner, October 2024
2. Lawrence Livermore National Laboratory Data Science Challenge Fellowship, August 2024